Sahaj Bhadja

(213) 284-4651 | bhadja@usc.edu | LinkedIn | GitHub | Portfolio

Summary

Software quality engineer with 2+ years validating cloud applications, APIs, and cross-platform systems. Experienced in Python, Linux, Git, test planning, automated testing, defect investigation, REST API validation, Docker, AWS, Azure, CI/CD, monitoring, and Agile delivery.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Built Python APIs and Azure SQL services for 500+ users, profiling indexed queries and backend execution paths to reduce production response latency from 1.2 seconds to 350 milliseconds.
- Validated 5,000+ monthly workflows through explicit states, retries, reconciliation, and failure isolation, tripling throughput while improving observability, deterministic recovery, reliability, maintainability, and systematic production anomaly diagnosis.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Developed and tested Flask REST services handling 10,000+ daily requests, validating secure file ingestion, authentication, status tracking, and error handling while resolving production defects affecting 150+ users.
- Created multithreaded processing for 1,000+ files with input validation, AWS RDS persistence, progress monitoring, and failure recovery, reducing turnaround 60% while maintaining reliable behavior under concurrent load.

Aasma Technology Solutions **India**
Software Developer Engineer – SecureSync *August 2023 – July 2024*
- Maintained 20+ automated unit and integration suites using Jenkins, PyTest, and Mockito, accelerating production defect investigation, release verification, and reliable validation across Windows, Linux, and macOS environments.
- Automated nightly VM workflows for builds, installation, integration testing, coverage reporting, and failure logs, improving cross-platform test repeatability while supporting root-cause analysis across globally distributed development teams.

Aasma Technology Solutions **India**
Full Stack Developer Intern – VibrantChem *May 2023 – August 2023*
- Validated eight microservices supporting authentication, role-based access, WebSocket messaging, and real-time synchronization across 2 million+ Azure SQL records through comprehensive automated functional, integration, and interface testing workflows.
- Automated Docker builds, functional tests, staged deployments, and production monitoring through Jenkins and Azure, reducing release time 50% while improving service reliability, deployment consistency, and operational visibility.

Cyber Security and Data Mining **India**
Data Analyst – Research Fellowship *May 2023 – August 2023*
- Built a robust malicious-URL detection pipeline over 650,000+ URLs using 21 behavioral indicators and machine-learning models, identifying phishing and malware patterns while achieving 96%+ classification accuracy.
- Evaluated Random Forest, Gaussian Naive Bayes, RNN, LSTM, Bi-LSTM, GRU, and CNN approaches, documenting model behavior and validating repeatable detection quality across diverse production-like security-focused test datasets.

Skills

Testing & Quality: automated test suites, build validation, input validation, functional testing, unit testing, integration testing, API testing, defect investigation, root-cause analysis, coverage reporting, quality reporting
Languages & Systems: Python, Java, JavaScript, TypeScript, SQL, Linux, Windows, macOS, object-oriented programming, multithreading, scripting, troubleshooting
Tools & Platforms: Git, GitHub, Jenkins, PyTest, Mockito, Docker, Kubernetes, virtual machines, AWS, Azure, S3, AWS RDS, Azure SQL, REST APIs, microservices
Delivery & Operations: CI/CD, Agile, Scrum, monitoring, logging, failure recovery, release verification, production support, documentation, cross-functional collaboration, cloud services, AI-assisted development

Projects

CPU-Scheduling Simulator | Live | *C++, React, JavaScript, Algorithms, Operating Systems*
- Built a C++ simulator implementing six scheduling algorithms with deterministic workloads, edge-case validation, Gantt visualization, and performance testing, reducing average turnaround time 25% across 1,000+ process configurations.

Chat2SQL | Code | *Java, TypeScript, SQL, PostgreSQL, MySQL, MongoDB, Docker*
- Built and validated a cross-platform workspace supporting 10+ databases with schema-aware query generation, structured execution, Docker-backed deployment, and modular Java and TypeScript services across Windows, macOS, and Linux.

FinNexus | Live | *TypeScript, Next.js, REST APIs, WebSockets, MongoDB, Kubernetes*
- Built a scalable platform supporting 50,000+ users and one million transactions, validating secure APIs, real-time WebSocket synchronization, MongoDB persistence, Kubernetes deployment, production monitoring, and automated release workflows.